

Model review and revisions

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I reviewed the model and calibration, including preferences, credit, demographic accounting, and the empirical targets.

- Returned to a Stone–Geary housing requirement that applies only to parents, retaining nonlinear benefits from children.
- Tested equivalence scales and changing housing shares.
- Rebuilt the first-birth housing evidence and national input measures.
- Revised adult entry and tested how purchase timing affects the calibration.

Household decisions

Overlapping generations of households differ in age, earnings, wealth, housing tenure, and family size.

- During fertile ages, households choose whether to attempt a birth; success becomes less likely with age.
- They choose consumption, saving, housing size, and whether to rent or own; renters pay rent, while owners make a down payment and face selling costs.
- Persistent earnings risk and borrowing limits shape saving; a first-birth fixed cost and taste shocks shape fertility.
- Households retire, face mortality risk, and value bequests.

Children and housing needs

The first child requires basic space at home; each additional child raises the goods needs of a larger family.

$$u(c, s; m) = \frac{\left[\frac{c^{\alpha_0} s^{1-\alpha_0}}{e(m)} \right]^{1-\sigma}}{1-\sigma} + \psi m^{1-\gamma},$$
$$e(m) = \left(\frac{2 + 0.7m}{2} \right)^{0.7}, \quad s = \eta_\tau [h - h_P \mathbf{1}\{m > 0\}],$$
$$\eta_R = 1, \quad \eta_O = \chi.$$

Here h is rooms, m is children at home, and $\tau \in \{R, O\}$ denotes tenure. Parents require $h > h_P$; χ raises housing services for owners. The equivalence scale $e(m)$ increases family consumption needs. Child benefits are zero at $m = 0$; ψ sets their level and γ controls diminishing marginal benefits.

Demographic and market equilibrium

Children leaving home and new adult households entering the economy are separate transitions.

- Children leave dependency independently; adult entry follows earlier births with a lag.
- Positive estates finance entrant assets; any remainder leaves the economy.
- Stationary calibration normalizes population to one: the house price satisfies birth renewal, and housing clearing determines H_0 .
- A balanced payroll tax finances equal retiree pensions.

Housing around the first birth

We found a mistake in the earlier regression and rebuilt the PSID event study using interview dates and household status before birth.



At years 3–4, rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Reference person or spouse before birth. Baseline: years $-3/-2$; person and year effects; survey weights; 95% pointwise intervals.

Externally set parameters

Parameter	Value	Source
Model period	4 years	
Entry and retirement ages	18, 66	
Risk aversion σ	2	
Equivalence scale (ω, ν)	(0.7, 0.7)	Scholz et al. (2006)
Consumption share α_0	0.733	CEX 2019–23, childless renters
Annual real interest rate	2%	Greaney et al. (2025)
Annual depreciation; property tax	1.416%; 1.060%	National housing inputs
Financed share ϕ	0.80	20% down payment
Selling cost ψ_s	6%	Greaney et al. (2025)
Owner house sizes	2, 4, 6, 8, 10 rooms	
Largest rental unit	6 rooms	Greaney et al. (2025)
Housing-supply elasticity	0.63	Baum-Snow and Han (2024)
Pension / working-household earnings	0.229	CPS ASEC, income 2007
Payroll tax	8.0%	Balances the pension budget
Bequest wealth shift θ_1	0.008	1% of median annual earnings

Parameters and target moments

2007 stationary economy: ten parameters, eleven moments.

Parameter	Interpretation	Main target moment	Target
<i>Saving and bequests</i>			
β	Patience	Wealth / earnings	6.927
θ_0	Bequest weight	Annual bequests / wealth (%)*	0.729
<i>Housing and tenure</i>			
χ	Owner housing premium	Ownership, ages 30–55 (%)	67.626
κ_T	Tenure tastes	Recent-parent ownership difference (pp)	12.761
h_P	Basic space for parents	First-birth rooms difference*	1.465
–	Additional moment	Mean rooms*	5.729
<i>Fertility</i>			
ψ	Child-benefit level	Completed fertility	2.100
ξ	First-birth cost	Childlessness, ages 40–44 (%)	19.828
κ_1	First-birth taste dispersion	Mean first-birth age	25.976
γ	Child-benefit curvature	One child among mothers (%)	21.366
κ_C	Later-birth taste dispersion	Children at age 25	0.810

Each parameter is paired with the moment it mainly moves; all ten are estimated jointly. *Recomputed: mean rooms (AHS), bequests (SCF), first-birth rooms (PSID).

Credit timing

A renter with financial wealth b and period income y who buys a home worth $Q = pH$:

$$c + (\delta + \tau_H) Q + b' = R(b - Q) + y.$$

Normally the buyer must have the 20% down payment at closing: $b \geq (1 - \phi)Q$, with $\phi = 0.8$.

The problem. A model period is four years. Requiring the down payment from wealth at the start of the period ignores saving within the period; counting all of it lets a buyer close with little equity and reach 20% only over the next four years. The rules differ in how much of the period's net saving $S = b' - (b - Q)$ counts:

$$b + \lambda S \geq (1 - \phi) Q.$$

- Soft, $\lambda = 1$: all of the period's saving counts.
- Quarter-saving, $\lambda = \frac{1}{4}$: a quarter of it counts, roughly one year of saving.
- Hard, $\lambda = 0$: only wealth brought into the period counts.

Stone–Geary calibration

2007 stationary economy; 80% mortgage financing

Moment	Target	Soft	Hard	Quarter-saving
Completed fertility	2.100	2.100	2.100	2.100
Childlessness, ages 40–44 (%)	19.828	20.222	20.413	20.546
One child among mothers, ages 40–44 (%)	21.366	21.264	20.907	20.844
Mean first-birth age (years)	25.976	26.032	26.007	26.057
Wealth / earnings	6.927	6.551	6.610	6.645
Annual child bequests / wealth (%)	0.729	0.685	0.684	0.685
Mean occupied rooms	5.729	5.957	6.241	6.131
Ownership, ages 30–55 (%)	67.626	66.527	60.878	64.572
First-birth rooms difference	1.465	1.324	1.063	1.167
Recent-parent ownership difference (pp)	12.761	11.825	10.125	11.683
Children at age 25 (capped at three)	0.810	0.528	0.524	0.521
Validation moments	Target	Soft	Hard	Quarter-saving
First births at age 30+ (%)	24.928	23.448	23.078	23.395
Older wealth/income p90/p50	3.516	2.968	2.861	2.937
Family rooms difference	0.385	0.304	0.321	0.316

Parameter estimates

Estimates under the three purchase rules; H_0 follows from housing clearing.

Parameter	Economic interpretation	Soft	Hard	Quarter-saving
β	Annual discount factor	0.967	0.967	0.968
χ	Owner housing-service premium	1.097	1.068	1.091
ξ	First-birth fixed utility cost	0.352	0.385	0.397
κ_1	First-birth taste dispersion	0.124	0.123	0.128
κ_C	Later-birth taste dispersion	0.363	0.350	0.348
θ_0	Bequest weight	0.106	0.106	0.109
γ	Child-benefit curvature	0.101	0.101	0.098
κ_T	Tenure-choice dispersion	0.014	0.014	0.015
ψ	Child-benefit level	0.178	0.172	0.174
h_P	Basic space requirement for parents (rooms)	2.504	2.562	2.492
H_0	Housing supply coefficient	6.757	7.546	7.289

More credit raises ownership but not births

Financed share 80% vs. 100%, house price held fixed

	80%	100%
Ownership, ages 30–55 (%)	64.6	74.0
Ownership, ages 18–21 (%)	10.5	31.4
First-birth probability, ages 18–21 (%)	25.5	25.8
Childlessness, ages 40–44 (%)	20.5	20.9
Completed fertility	2.10	2.09

First births rise slightly at 18–21 but fall at later ages, so childlessness rises and completed fertility falls slightly.

This comparison holds the house price fixed. In dated transitions where the house price adjusts, a temporary move to 100% financing at the recalibrated hard and quarter-saving points lowers first births by about 0.5% in the first period.

Why the response is small: what I have checked

Credit raises births only if a childless renter delays a child because, after a birth, renting would leave the family short of space and the down payment blocks buying. Three things must hold:

1. **After a birth, renters are short of space.** Yes: about 80% of new parents who keep renting are at the six-room cap.
2. **Being short of space is costly.** Not very: one room short costs about 2% of consumption for one period under Cobb–Douglas preferences.
3. **Credit makes the space cheaper.** No: a room costs the same per period rented or owned. Credit gives access to homes above six rooms, not cheaper space, and the ownership premium χ is the same for parents and non-parents.

So credit removes an obstacle that exists but might be small. I have not yet shown how much of the small response this accounts for.

What I want to try next

- Selling costs: check whether the cost of trading up after a birth is what keeps credit from reaching fertility, and whether the 6% seller-paid cost is the right convention.
- Small owner homes: young buyers may lock into homes too small for a family.
- A housing requirement that grows with the number of children.
- Improve the calibration.
- Reconsider the utility specification.